System Driven Hardware Design

- Heart Beat and Push Button Interrupt -

International Master of Science

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Agenda

- **Get the Toolchain Running**
- Getting Familiar with FreeSoc 2
- User LED - Heart Beat
- External LED - Heart Beat
- Push-Button Turn On Turn Off

Get the Toolchain Running

→ Instlla PSoC Creator: https://www.cypress.com/products/psoc-software
→ See the HowTo flash Video of the Programmer in this Moodle section
→ Check https://www.sparkfun.com/products/13714 for more information
→ Check the Folder in Moodle for FreeSoc 2

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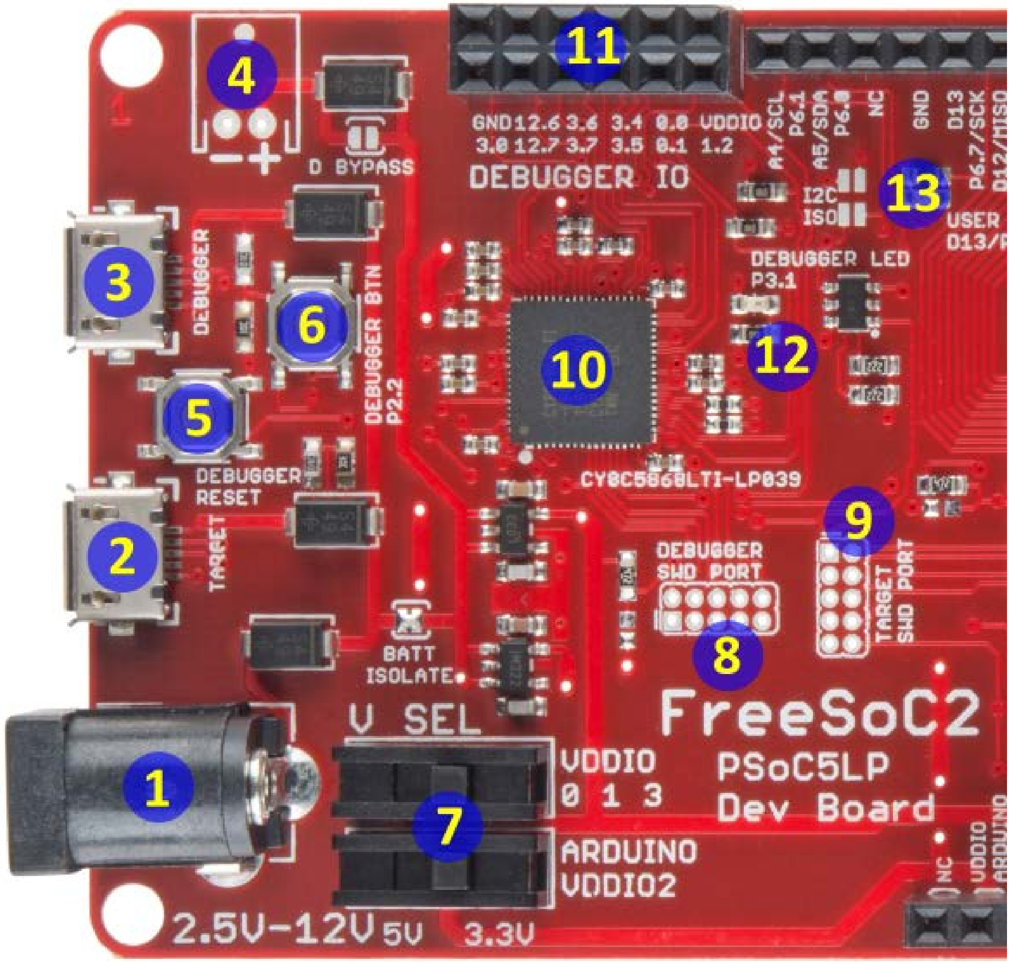
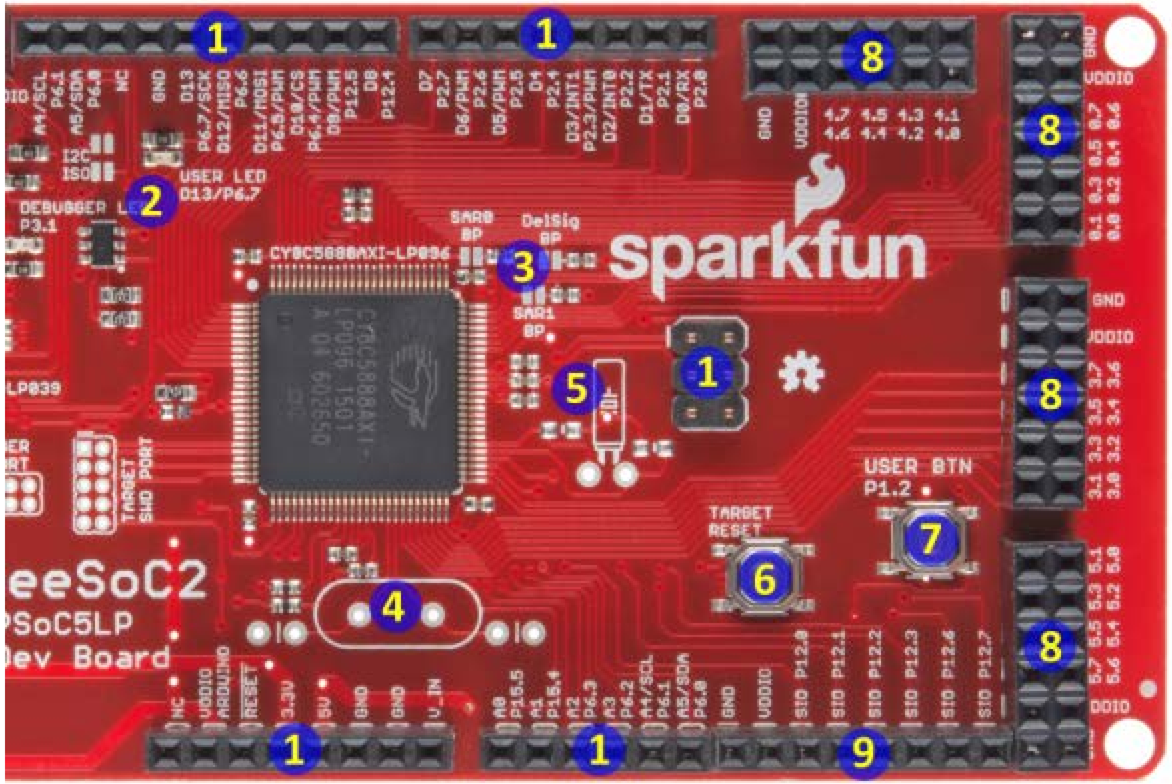
Getting Familiar with FreeSoc 2



Getting Familiar with FreeSoc 2



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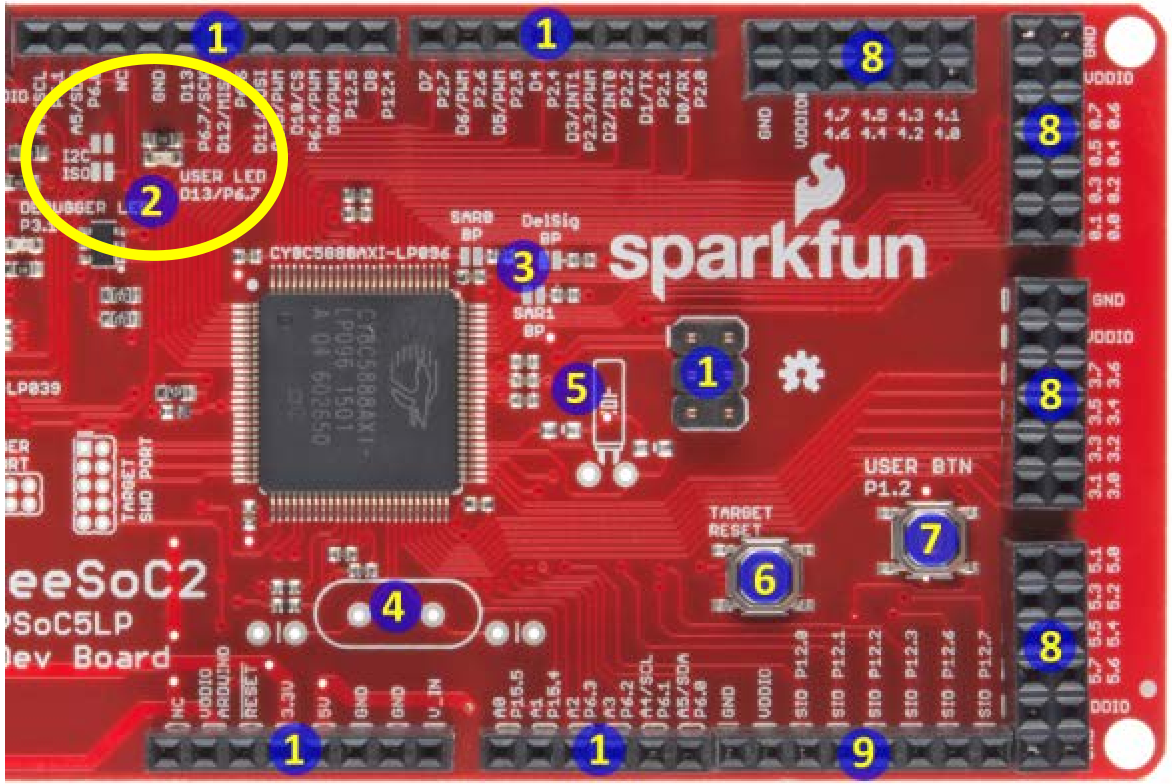
USER LED
D13/P6.7
sparkfun
USER BTN
P1.2
TARGET
RESET
CY8C5888AXI-LP096

User LED – Heart Beat

A very good starting point is to do the following tutorial
https://github.com/open-storm/docs.open-storm.org/wiki/Tutorial:-Getting-started-with-PSoC-Creator-%7C-Blinking-an-LED

Adapt it to FreeSoc2:
- Your device: CY8C5888AXI-LP096
- LED is the “User LED” on the board (see Board and schematic, how to use it)
- Assign the correct pin

User LED – Heart Beat

- Make a copy of your Design
- Adapt the copy to the following: use an external LED on the Bread Board
- Question to be cleared:
    - What's the forward voltage and current of the external LED?
    - Which voltage provides a digital PSoC pin and which current can it source and sink? (check datasheets for this)
    - How should an external circuit look like, in order to drive the external LED and to turn it on by an PSoC pin?

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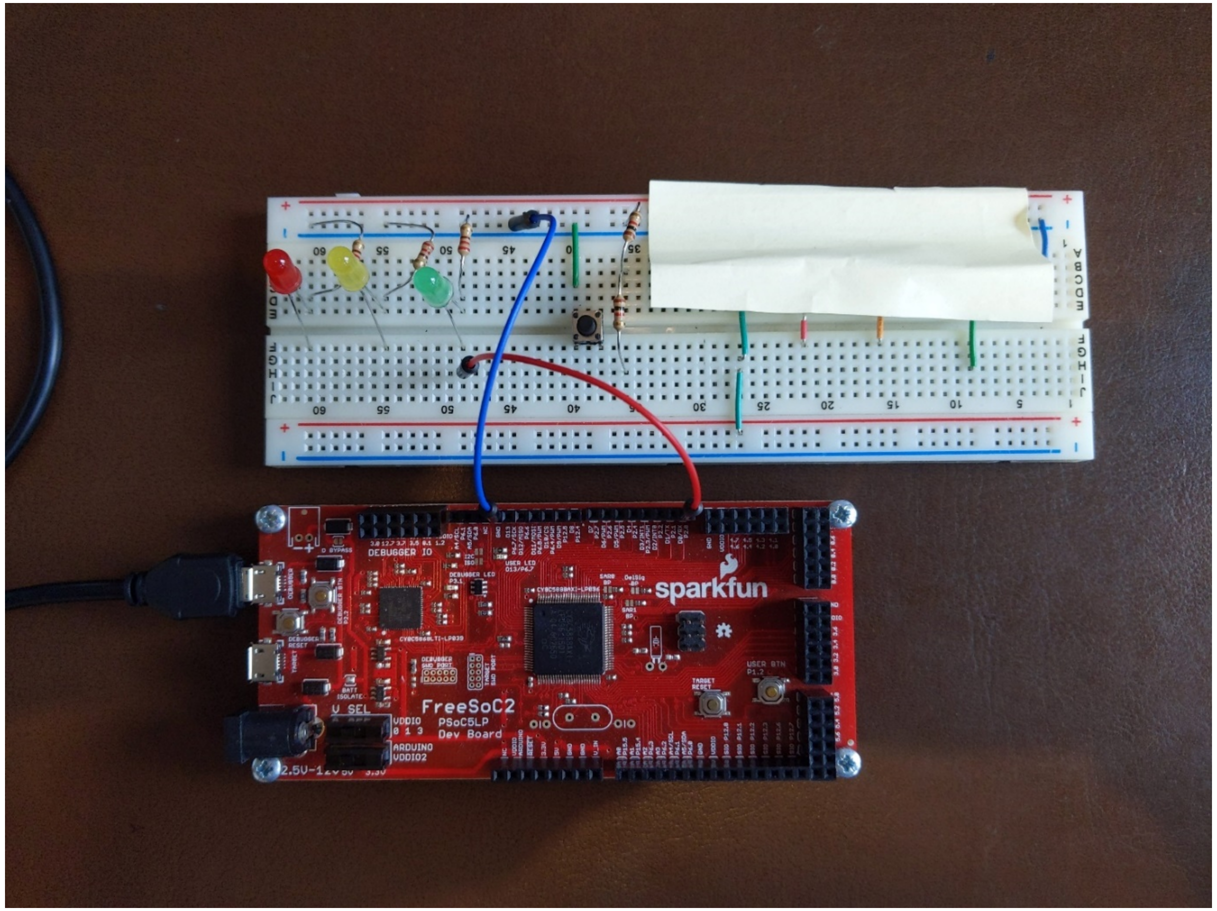
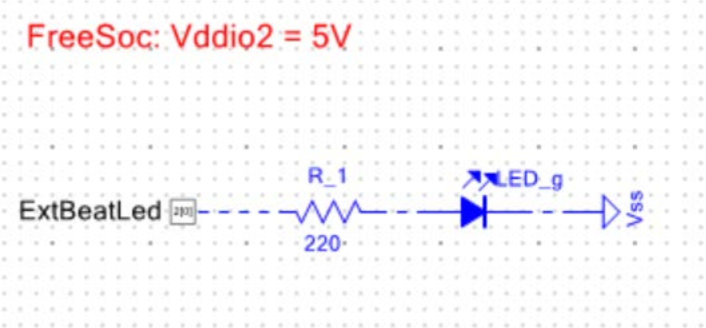
External LED - Heart Beat





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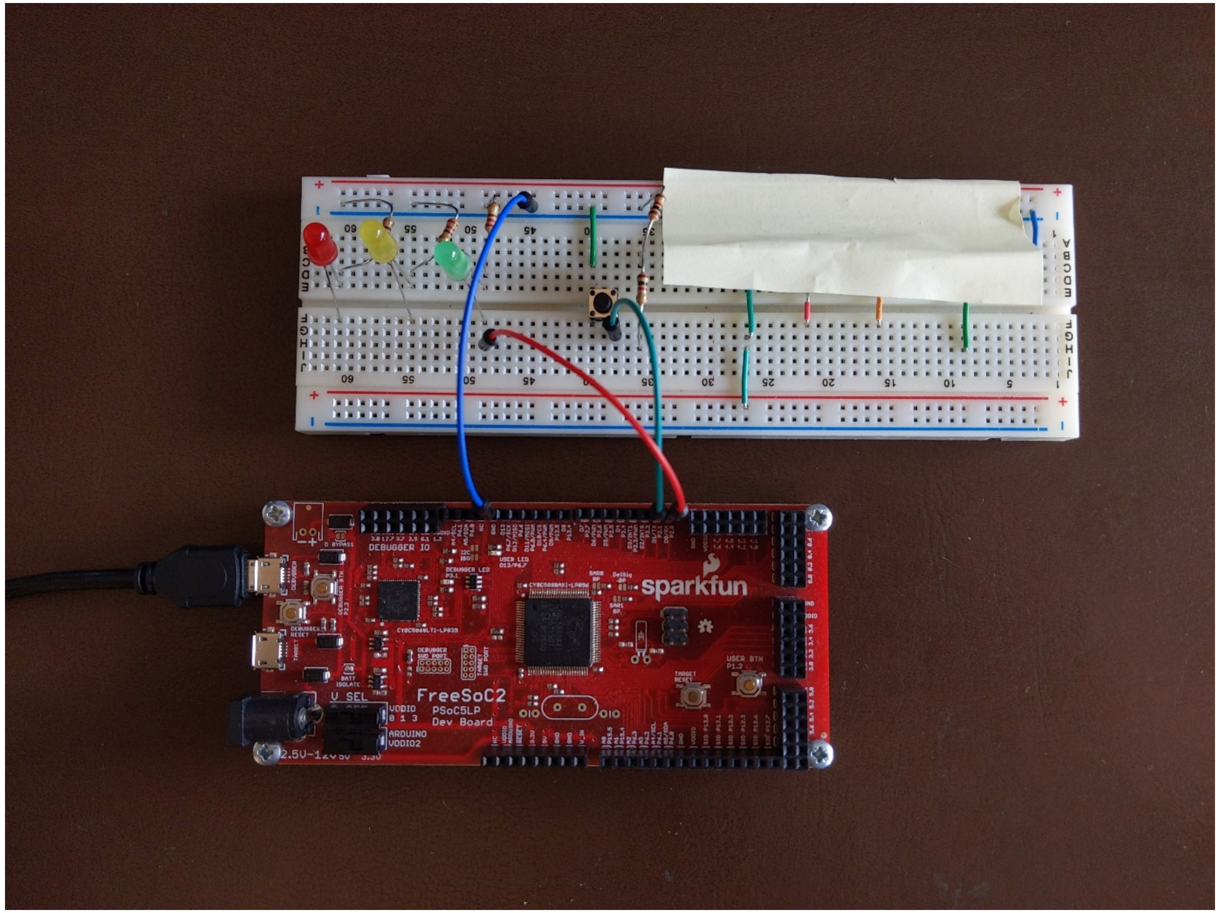
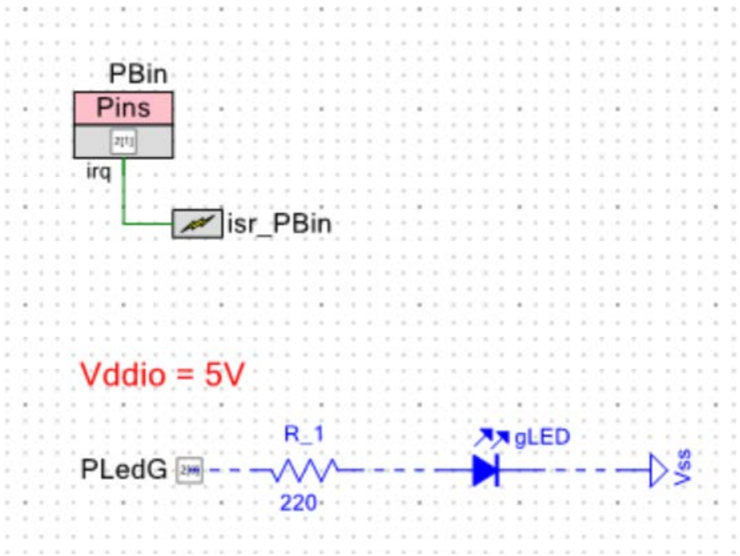
Push-Button Turn On Turn Off



Top Schematic



Your Exercise for the next week

1. Implement a switch debouncing
2. Implement a 3 Bitt Counter
   - Pb counts up and 3 LEDS (red – yellow – green) shows the counting